

Total No. of Questions : 4]

SEAT No. :

PD93

[Total No. of Pages : 1

[6410]-414

T.E. (Chemical) (Insem)

PROCESS INSTRUMENTATION AND CONTROL
(2019 Pattern) (Semester - II) (Elective - II) (309351B)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Figures to the right indicate full marks.
- 2) You are advised to attempt not more than two questions.
- 3) Neat diagrams must be drawn whenever necessary.
- 4) Use of logarithmic tables, slide rule, Miller charts, electronic pocket calculator and system table is allowed.
- 5) Assume suitable data, if necessary.

- Q1) a)** Explain functional elements of instruments in detail. [7]
b) Differentiate between analog and digital instrument. [8]

OR

- Q2) a)** Explain the importance of instrumentation in process industries. [7]
b) Explain static and dynamic characteristics of a measuring instruments. [8]

- Q3) a)** Explain the working of the gas filled thermometer with the help of neat diagram. [7]
b) Explain with Principle, diagram, construction and working of Thermocouple. [8]

OR

- Q4) a)** Write short note on Helix bimetal type mechanical temperature sensors. [7]
b) Explain optical pyrometer with the help of neat diagram. [8]

